

Exact discrete-time stability analysis of multi-DOF haptic rendering: Impact of multi-rate, time-delay, and mechanical parameters

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Abstract

Previous studies on the stability analysis of haptic devices have predominantly focused on single-DOF devices, thereby limiting attention to multi-DOF devices, particularly those employing multi-rate sampling schemes. In this paper, we introduce a formulation for the coupled dynamics between the haptic device and the virtual environment for a multi-DOF haptic device controlled using a dual-rate sampling scheme. Subsequently, we analyze its stability through the application of a dynamic decoupling strategy within an exact discrete-time state-space framework while the device is engaged in rendering a virtual wall along one of its operational space coordinates. Furthermore, we explore how the combined influence of the dual-rate sampling approach, time delay, and the mechanical design affects the stability boundaries of the multi-DOF haptic device at a fixed workspace location as well as within the entire usable workspace. Additionally, we utilize a model-order reduction (MOR) framework to simplify the determination of the device's stability limits, irrespective of the specific combinations of time delay and sampling rates employed.

Keywords

Haptic interface, dual-rate sampling, multi-DOF, stability, *Z-width*, model-order reduction

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Introduction

Human-machine interfaces (HMIs), such as head-mounted displays, headsets, body suits, and haptic devices, form an integral part of teleoperated and extended reality systems by enabling information exchange between the rendered environment and the user.¹ Among these HMIs, haptic devices facilitate bidirectional interaction through the sense of touch, conveying information about the geometrical, inertial, and material properties of remote or virtual objects.² The incorporation of haptic-based feedback in virtual and remote interactions offers significant benefits, such as reducing cutting forces in remote or virtual surgery,³ assisting sound localization in hearing-impaired individuals,⁴ and enhancing the sense of immersion.⁵

Haptic devices have been designed across a spectrum, spanning from expansive full-body exoskeletons to compact fingertip-sized tactile devices.^{6,7} However, the most widely adopted category in both research and commercial spheres continues to be the grounded kinesthetic devices with force feedback, known as *impedance-type* devices.⁸ For a realistic portrayal of virtual or remote environments, these devices are

required to possess a wider mechanical bandwidth, often referred to as the *Z-width*.^{9,10} From a structural standpoint, meeting this demand necessitates that haptic devices exhibit minimal inertia and high structural stiffness.¹¹ Haptic devices utilizing a parallel kinematic architecture excel in meeting these prerequisites, making them prevalent in various applications. For instance, Koehler et al.¹² have very recently developed a 5-DOF parallel haptic device with compliant linkages, while Williams et al.¹³ have designed a finger-mounted 4-DOF device using origami principles. Simpler multi-DOF parallel haptic devices, such as the 2-DOF 5 bar mechanism capable of providing force feedback within a planar workspace, have found applications in various fields.¹⁴ Such devices have

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been utilized independently or as part of a more complex mechanism.^{15,16}

It is known that the performance of haptic devices is significantly restricted, especially in emulating rigid environments, as reflected in their finite *Z-width*. In addition to the device architecture, the constrained *Z-width* of haptic devices is attributed to control factors such as sampling rate, time delay, quantization, etc., stemming from the digital implementation of haptic rendering.¹⁷ The sampling rate, in particular, restricts the maximum *Z-width* due to *energy leaks*,¹⁸ with higher rates worsening the velocity and acceleration estimation based on discrete and quantized position measurements.^{9,19} Discrete sampling's impact on *Z-width* is exacerbated by time delay.²⁰ Additionally, the damping exhibited by haptic devices, whether from inherent joint friction or other mechanisms, influences the *Z-width*, with higher physical damping corresponding to a wider *Z-width*.^{21–23}

One approach to enhance the *Z-width* relies on increasing the physical damping of haptic devices. Various strategies, including mechanical, electrical, and digital methods, have been proposed for this purpose.^{9,22,24} However, these approaches impact the lower threshold of the *Z-width* and thereby degrade the device transparency. An alternative method to improve the *Z-width* while preserving transparency involves the utilization of multi-rate control strategies.^{25,26} One such multi-rate strategy, referred to as dual-rate sampling, decouples the rendering of virtual environment components into velocity-dependent and velocity-independent loops.²⁶ This decoupling allows the two loops to be sampled at appropriate rates independently and ensures emulation of high virtual stiffness without compromising estimated velocity. For multi-DOF haptic rendering, where maintaining higher sampling rates is challenging due to the involvement of multiple sensors and actuators, as well as the increased computational load introduced by tasks such as estimation of forward kinematics, force-torque mapping, and virtual environment implementation, such multi-rate strategies have proven particularly beneficial.²⁷

Ensuring the stability of haptic devices is crucial in virtual or remote interactions where a human operator is directly involved. Stability analysis of haptic devices thus remains a prominent and ongoing area of research within the field of haptics. Previous studies have explored the impact of factors like physical damping, sampling rate, position quantization, velocity filtering, Coulomb friction, and time delay on single-DOF devices, utilizing frameworks such as passivity analysis, network theory, and linear control methods.^{17,19,21,28–30}

Linear control theory has been particularly utilized for stability analysis of haptic devices under different conditions and considerations. For instance, Gil et al.^{20,31} applied the Routh-Hurwitz criterion to assess a single-DOF haptic device's stability,

considering factors like virtual damping and time delay. In contrast, Mashayekhi et al.^{23,32} derived closed-form stability equations without constraints on these parameters. Other studies focused on the impact of varying and uncertain time delays,^{33,34} virtual mass rendering,³⁵ and fractional-order differentiation³⁶ on the stability of single-DOF haptic devices.

Recent works on haptic device stability and performance enhancement include.^{37–44} For instance, Nomberg and Nisky^{37,38} analyzed human-in-loop stability, incorporating human operator dynamics and time delay using a root tracking methodology. Oliver et al.⁴⁰ established stability boundaries for virtual environment rendering based on Hunt-Crossley models, while Chen et al.⁴¹ explored haptic device stability using fuzzy-passivity control. Mashayekhi et al.⁴² developed a describing function (DF) for the Gaussian friction model, enabling the analysis of stability for a 1-DOF haptic device. The developed DF accounts for nonlinear effects such as stiction and the Stribeck effect, allowing prediction of stability boundaries without time delay and virtual damping constraints. Their simulation and experimental results indicate an 80% improvement in prediction accuracy when employing the Gaussian DF compared to scenarios without its use. Pecly and Hashtrudi-Zaad⁴³ proposed three methods for computing low-latency analog position signals in a 1-DOF haptic device. These methods leverage sensor fusion of encoder and rate gyro signals, achieving over 98% accuracy compared to sampled encoder signals. In experimental tests, the proposed analog methods outperformed digital implementations, providing a wider virtual stiffness range (up to 400%) with enhanced fidelity during virtual wall rendering. In Dinc et al.⁴⁴ a less conservative approach is introduced for haptic interaction to achieve a wider impedance range and high transparency. By monitoring energy generation in each pressing-releasing cycle of haptic interaction with a virtual environment, adaptive damping is injected using low-frequency control actions to regulate energy until depleted, ensuring system convergence. Notably, all the aforementioned studies utilized a uniform-rate sampling scheme for virtual environment rendering and are limited to single-DOF haptic rendering.

There is a limited body of research addressing the stability aspects of multi-rate control systems for haptic rendering.^{25,26,45–47} For instance, Lee and Lee,²⁵ derived stability conditions for their multi-rate controller using Taylor series equivalents and the Routh-Hurwitz criterion. Barbagli et al.⁴⁵ used the lifting technique to investigate the stability of their multi-rate system. Fotoohi et al.⁴⁷ compared distributed and centralized multi-rate haptic control schemes in terms of stability and fidelity. In our previous research,^{26,48–50} the focus was on analyzing the stability of single-DOF haptic devices utilizing the dual-

rate sampling strategy. For instance, in Koul et al.,²⁶ an approximate analysis combined Taylor series approximation with the Routh-Hurwitz criterion to establish linear stability conditions for non-delayed dual-rate haptic rendering. This study was expanded upon in Ganiny et al.,⁴⁹ where various exact discrete-time and approximate continuous-time methods were used to assess the stability and identify the most conservative stability boundaries. In Ganiny et al.,⁴⁸ the combined impact of time delay and dual-rate sampling on the stability margins of single-DOF haptic devices was examined. Furthermore, our work in Ganiny et al.^{48,50} demonstrated simplifying the stability analysis of time-delayed dual-rate haptic rendering with MOR techniques.

Despite the proliferation of multi-DOF haptic devices in various applications, research on their performance and stability aspects has received significantly less attention. Typically, stability investigations for such devices have been experimental, often focusing on the motion of the device around a single active joint while constraining the other joints physically.^{20,23,31,35} This experimental approach has been employed to validate theoretical findings developed for single-DOF devices using commercially available multi-DOF haptic devices or robots. However, Amin et al.^{51,52} have developed a framework for modeling and analyzing the stability of multi-DOF haptic devices, treating them as either rigid or flexible systems. In their subsequent work,¹⁰ they explored the role of devices mechanical properties on the *Z-width* of multi-DOF haptic devices. Additionally, Mashayekhi et al.⁵³ recently developed a simulation-based framework to simplify multi-DOF haptic device dynamics into single-DOF systems. Notably, these studies are limited to haptics controllers that employ a uniform-rate sampling strategy for rendering virtual environments. Furthermore, none of these studies have attempted to simplify the determination of stability boundaries through techniques such as MOR or other equivalent frameworks.

The review presented above highlights a significant disparity in the attention given to the stability analysis of multi-DOF haptic devices as compared to their single-DOF counterparts. In particular, there is a lack of studies that explore the stability of multi-DOF devices employing unconventional control methods, especially those that utilize multi-rate sampling strategies. This scarcity arises due to the increased complexity in formulating device kinematics, governing dynamics of the coupled device-virtual environment system, and simplifying the hybrid multi-rate control loops for multi-DOF haptic systems to a single time-domain representation and a single sample rate. Dynamic coupling among various DOFs in multi-DOF haptic devices adds complexity to stability examination, especially in directions where virtual environments are implemented. Factors like time delay further complicate the analysis of such devices.

In view of the aforesaid research gaps, the current research work is motivated by the imperative to address the stability of multi-DOF haptic devices utilizing the dual-rate sampling strategy for virtual environment implementation. Our primary objective is to elucidate the establishment of governing dynamics of the coupled device-virtual environment system, modeled as a complex multiloop and multi-rate system, with a particular emphasis on understanding the decoupled dynamics within the device's DOFs. This study aims to determine stability boundaries for scenarios involving the rendering of virtual walls in a specific direction, employing an exact discrete-time stability analysis within the state-space formulation. Furthermore, we endeavor to simplify the determination of these stability boundaries by utilizing a stability-preserving MOR technique, especially in cases where higher time delays and sampling rates are involved. Additionally, we seek to evaluate how the mechanical design of the device, particularly parameters like the base link and other link lengths, interacts with the dual-rate sampling strategy, thereby influencing the maximum renderable virtual stiffness. Finally, we assess the impact of the dual-rate sampling scheme on the maximum renderable virtual stiffness across different workspace locations.

The paper's subsequent sections are organized as follows: In Section 2, we revisit the previously proposed dual-rate sampling scheme, originally intended for single-DOF haptic devices and detail its multi-DOF extension. In the subsequent section, we provide a comprehensive description of the system under consideration for our analysis. Sections 4 and 5 present the kinematic and dynamic formulations of the device, respectively, with a focus on decoupled and linearized dynamics. Section 6 is dedicated to the exact discrete-time stability analysis of the multi-DOF device under consideration. The simplification of the stability assessment using balanced truncation based MOR is detailed in Section 7. In Section 8, we present and discuss the results of the stability analysis. Finally, the concluding remarks and potential directions for future work are provided in Section 9.

Dual-rate sampling scheme

In this section, we describe the multi-DOF variant of the dual-rate haptic control scheme proposed by Koul et al.²⁶ for single-DOF systems. We begin by highlighting the limitations of the conventional open-loop impedance control scheme, referred to herein as the uniform-rate control scheme, when rendering virtual walls consisting of stiffness and damping components at higher controller sampling rates. This underscores the motivation behind utilizing dual-rate sampling scheme for haptic rendering. The subsequent portion of this section then describes the control architecture of the dual-rate haptics controller.

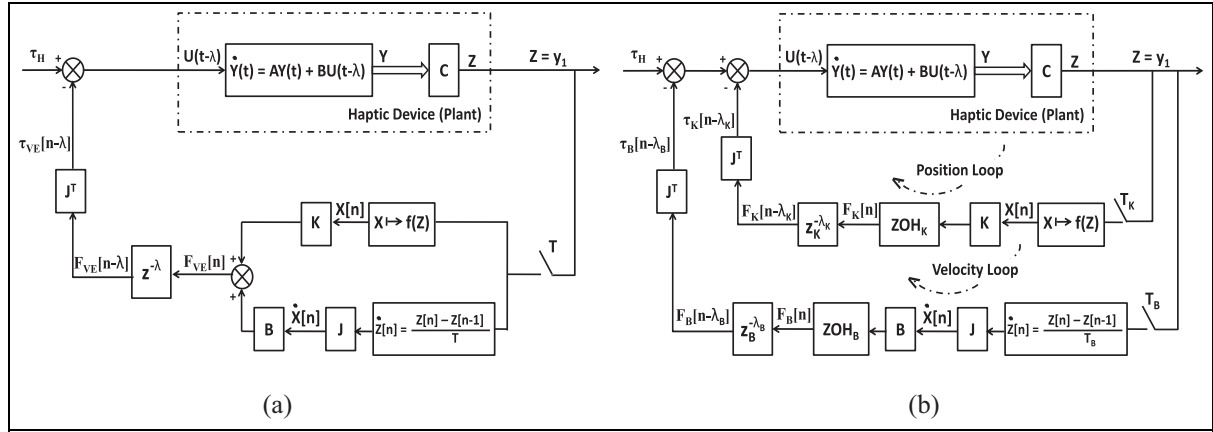


Figure 1. Schematic representation of uniform rate and dual-rate haptics controllers for multi-DOF haptic rendering: (a) conventional uniform-rate haptics controller, and (b) dual-rate haptics controller.

Limitations of uniform-rate sampling scheme

The block diagram of a multi-DOF haptic device utilizing a uniform-rate sampling scheme is shown in Figure 1(a), as employed for the implementation of virtual walls, which are modeled as parallel spring-damper systems. In this context, we consider the common practice of using optical encoders for position sensing and estimating velocity from discrete position samples using the finite difference method (FDM).^{9,31,54}

Virtual wall rendering is a benchmark problem in haptics. Essentially, a virtual wall represents a constrained surface that provides significant impedance upon contact and exhibits zero impedance when not in contact with the device's avatar. When engaging with a virtual wall, a haptic device provides feedback forces corresponding to the wall's impedance, which typically depends on the stiffness (K) and damping (B) of the wall.^{19,23}

In the typical uniform-rate haptic control scheme, the device's state variables, usually representing joint positions in the case of a multi-DOF device, are sampled at a uniform rate, denoted as $f = 1/T$. The joint velocities are determined using FDM, where the difference between consecutively sampled joint position values is taken and then scaled by the inverse of the sampling period, T . The end-effector position, denoted as $\mathbf{X}[n]$, in the operational space where the virtual walls are rendered, is determined using the device's forward kinematic mapping $\mathbf{X} \mapsto f(\mathbf{Z})$, and the vector of end-effector velocities, represented as $\dot{\mathbf{X}}[n]$, is determined by pre-multiplying the vector of joint velocities, $\dot{\mathbf{Z}}[n]$, with the device's configuration-dependent Jacobian matrix, $\mathbf{J}$. The end-effector position and velocity components are subsequently multiplied by the stiffness matrix $\mathbf{K}$ and damping matrix $\mathbf{B}$, which encompass the virtual wall stiffness and damping coefficients of the rendered virtual walls, respectively, to estimate the feedback forces. For a 2-DOF haptic interface, the resulting feedback force due to motion within the virtual walls is given by:

$$\mathbf{F}_{VE} = \mathbf{K}\Delta\mathbf{X} + \mathbf{B}\dot{\mathbf{X}} \quad (1)$$

where, $\mathbf{F}_{VE}$, $\Delta\mathbf{X}$, and $\dot{\mathbf{X}}$ are the vectors of feedback forces, wall penetrations, and wall contact velocities, respectively. These vectors and the matrices $\mathbf{K}$ and $\mathbf{B}$ are defined as:

$$\mathbf{F}_{VE} = \begin{Bmatrix} f_x \\ f_y \end{Bmatrix}; \quad \Delta\mathbf{X} = \begin{Bmatrix} \Delta_x \\ \Delta_y \end{Bmatrix}; \quad \dot{\mathbf{X}} = \begin{Bmatrix} \dot{x} \\ \dot{y} \end{Bmatrix}$$

$$\mathbf{K} = \begin{bmatrix} K_x & 0 \\ 0 & K_y \end{bmatrix}; \quad \text{and,} \quad \mathbf{B} = \begin{bmatrix} B_x & 0 \\ 0 & B_y \end{bmatrix}$$

where, the subscripts x , and y refer to the X and Y operational space coordinate axes, respectively. The vector of feedback forces, $\mathbf{F}_{VE}$, is transformed to the vector of joint torques, τ_{VE} , by utilizing the transposed Jacobian matrix, $\mathbf{J}^T$. Moreover, due to factors such as wall contact estimation, communication, filtering, and so on, there is usually a time delay, denoted here as λ , within the control loop.

In this control scheme, the joint velocities are ubiquitously determined based on FDM, specifically the backward difference method, based on the joint angle measurements from optical encoders. However, joint velocity estimates obtained through the FDM often suffer from significant resolution limitations, especially at higher controller sampling rates.^{9,29} The noise in joint velocity estimates propagates to the operational space, affecting the estimation of end-effector velocity components. As a result, the noise introduced by the FDM significantly restricts the *Z-width* that the device can achieve, and at higher sampling rates, the noise in velocity estimates prohibits the stable implementation of any virtual damping.^{9,31} To address this issue, a common practice is to incorporate a low-pass filter in conjunction with the FDM.^{9,17,54} However, the use of low-pass filters introduces a time delay in velocity estimates, which, in turn, imposes an additional constraint on the

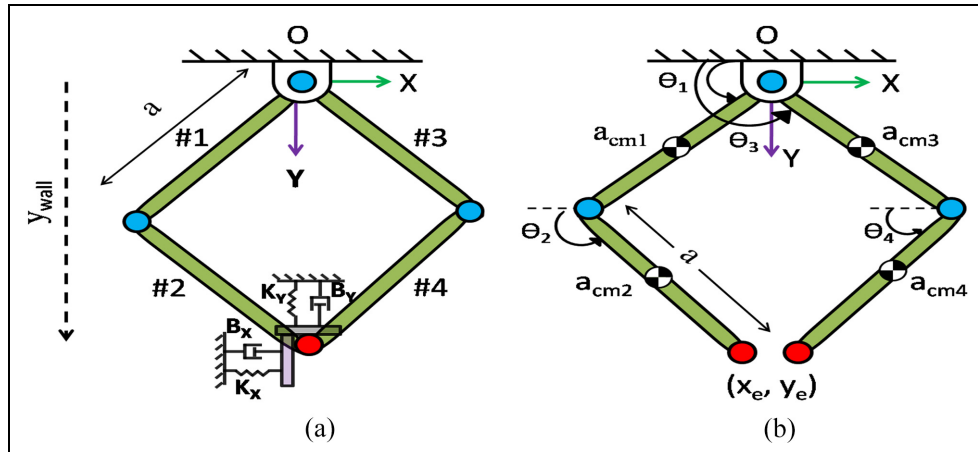


Figure 2. A schematic representation and kinematic diagram of a 2-DOF haptic device: (a) schematic representation of haptic rendering along two operational space directions, and (b) kinematic diagram of the 2-DOF haptic device.

achievable *Z-width* of the haptic device. The degree of low-pass filtering employed and, consequently, the attainable *Z-width* depend on the trade-off between the noise induced by the FDM and the time delay caused by filtering.²⁹ This factor has remained one of the fundamental challenges preventing the use higher sampling rates for uniform-rate haptic rendering, despite the fact that higher sampling rates are associated with broader *Z-widths*.^{9,19,31} Furthermore, this limitation restricts the utility of the uniform-rate control scheme to soft model simulations and rendering. It is precisely this limitation of the uniform-rate control scheme that motivates the adoption of a dual-rate control scheme to achieve high stiffness (wider *Z-width*) at higher sampling rates.²⁶

Architecture of dual-rate sampling scheme

The architecture of the dual-rate sampling scheme for multi-DOF haptic rendering is depicted in Figure 1(b). The issues exhibited by the uniform-rate control scheme at higher sampling rates, owing to noisy velocity estimates from encoder measurements, are addressed in the dual-rate sampling scheme by sampling the device state variables at two distinct rates, rather than a single rate that is employed in uniform-rate sampling scheme. In this dual-rate sampling approach, the virtual walls are rendered in a multi-loop implementation wherein the stiffness and damping parts of the virtual walls are decoupled and rendered separately at distinct sampling rates. These two loops are known as the position or stiffness loop and the velocity or damping loop, respectively. Specifically, for implementing the virtual stiffness components of the virtual walls the device joint positions are sampled at higher rates, denoted as $f_K = 1/T_K$, so that high stiffness can be implemented. Conversely, for the implementation of damping coefficients, relatively lower rates, denoted as $f_B = 1/T_B$ ($f_K > f_B$), are used for sampling joint positions and

determining joint velocity from FDM. This approach ensures that noise in the estimated velocity remains minimal, allowing for stable implementation of the damping coefficients. Additionally, the FDM is often cascaded with a low-pass filter to attenuate any high-frequency noise in velocity estimates. The filter characteristics are selected to balance noise attenuation and the induced delay from filtering, ensuring acceptable performance.

System description and assumptions

This paper focuses on investigating a non-redundant pantograph mechanism, specifically a 2-DOF planar 5 bar linkage device, for the purpose of formulating dynamics and analyzing stability of multi-DOF haptic devices operating under the dual-rate sampling strategy. This particular device possesses a parallel kinematic architecture that yields high structural stiffness and low inertia. As mentioned earlier, these characteristics are essential prerequisites for enabling haptic devices to effectively simulate high-impedance virtual environments. In Figure 2(a), the kinematic diagram of the device is presented, showing its key geometrical attributes, base coordinate system, and link numbering, among other details. Additionally, this diagram depicts the device end-effector rendering virtual walls along the two operational space directions *X* and *Y*.

The device consists of a pair of active links actuated by DC motors at the base end, along with two passive links at the distal end where the end-effector is attached. The device is fundamentally a congruent pantograph mechanism, as the non-grounded links are all of equal length and the ground link has a length of zero. This kinematic configuration leads to a larger workspace without singularities, in comparison to architectures with a non-zero base length, and generates a wider range of passive forces.⁵⁵ The link length is denoted by *a* and the mass of each link is

designated as m_i , where the subscript $i = 1, 2, 3, 4$ refers to the link number.

The flexibility of the links and the compliance resulting from the motor transmission is assumed to be negligible. Accordingly the spatial configuration of the links is well described by their corresponding joint angles and velocities in the joint space coordinates. The active joint angles are measured by the integrated optical encoders on the DC motors while the passive joint angles are calculated based on the device geometry and symmetry considerations. The joint angles are considered as the generalized coordinates for the dynamics formulation reported in the subsequent section. We express the vector of generalized coordinates as $\Theta \in \mathbb{R}^{4 \times 1}$, and denote the minimum set of generalized coordinates that is, the set of active joint angles by $\Theta_{ms} \in \mathbb{R}^{2 \times 1}$. These joint angle vectors are given by $\Theta_{ms} = [\theta_1 \ \theta_2 \ \theta_3 \ \theta_4]^T$ and $\Theta_{ms} = [\theta_1 \ \theta_2]^T$ respectively. The operational space coordinates are measured with reference to the X-Y coordinate system aligned with the base of the device. The position of the end-effector in the operational space is denoted as $\mathbf{X} = [x_e \ y_e]^T$ and the same is related to the generalized coordinates that is, the joint angles Θ via the forward kinematic equations. Conversely, the inverse kinematic equations define the mapping from the operational space back to the joint space.

As is usually the case, we assume that the active joint velocities are determined using FDM. At the j^{th} sampling instant the two active joint velocities, $\dot{\theta}_1$ and $\dot{\theta}_3$, are thereby given as

$$\begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_3 \end{Bmatrix} = \frac{1}{T_B} \begin{bmatrix} \theta_1[jT_B] & \theta_1[(j-1)T_B] \\ \theta_3[jT_B] & \theta_3[(j-1)T_B] \end{bmatrix} \begin{Bmatrix} 1 \\ -1 \end{Bmatrix} \quad (2)$$

Additionally, the end-effector velocity $\dot{\mathbf{X}}$ is obtained by mapping the joint velocities to the operational space through the Jacobian matrix $\mathbf{J}(\dot{\mathbf{X}} = \mathbf{J}\dot{\Theta})$. Moreover, it is assumed that any propagation of quantization in joint angles and their derivatives, resulting from the use of optical encoders, is negligible in operational space coordinates.⁵⁶

The device renders the two virtual walls characterized by parameters B_x , K_x and B_y , K_y along the two operational space viz. X and Y directions, respectively. It is assumed that the device remains horizontally oriented during interactions, and as such, gravity has no impact on its dynamics. The sampling periods of the position and the velocity loops are denoted by T_K and T_B , respectively. The time delay in the controller is denoted by λ , and the non-dimensional time delays in the position and velocity loops are designated by λ_K and λ_B .

Kinematics formulation

The foremost step in analyzing the stability of a multi-DOF haptic device involves establishing its

governing dynamic equations. This, in turn, necessitates the formulation of the device's forward and inverse kinematics, Jacobian matrix, etc. Forward kinematics maps joint space motion, Θ or Θ_{ms} , to end effector motion, $\mathbf{X}$, which can be expressed as $\mathbf{X} \mapsto f(\Theta)$ or $f(\Theta_{ms})$. The knowledge of the forward kinematic equations is essential because haptic devices usually render virtual walls within the operational space, while the position sensors are localized at the device joints. Furthermore, detecting wall contacts and estimating interaction forces based on wall impedance are only possible when the forward kinematics of the device are known. Conversely, inverse kinematic equations link desired end-effector positions $\mathbf{X}$ in operational space to joint angles, Θ or Θ_{ms} , expressed as Θ or $\Theta_{ms} \mapsto f(\mathbf{X})$. In what follows, we will formulate the kinematics of the 2-DOF haptic device under consideration using straightforward application of coordinate geometry and trigonometry.

The forward kinematic equations of the 2-DOF haptic device can be obtained by assuming that the device is cut into two serial chains at the end-effector as is illustrated in Figure 2(b).

In terms of generalized coordinates, Θ , the forward kinematic equations can thus be expressed as:

$$\begin{Bmatrix} x_e \\ y_e \end{Bmatrix} = a \begin{bmatrix} \cos \theta_1 & \cos \theta_2 \\ \sin \theta_1 & \sin \theta_2 \end{bmatrix} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} = a \begin{bmatrix} \cos \theta_3 & \cos \theta_4 \\ \sin \theta_3 & \sin \theta_4 \end{bmatrix} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} \quad (3)$$

These kinematic equations are helpful in evaluating the mapping between the generalized coordinates, Θ , and the minimum set of generalized coordinates, Θ_{ms} , as will be shown in the subsequent sections.

Since the position sensors on the device solely measure the active joint angles, it becomes essential to express the forward kinematic equations in terms of Θ_{ms} . With this consideration, the following constraint equations can be obtained:

$$\begin{aligned} (x_e - a \cos \theta_1)^2 + (y_e - a \sin \theta_1)^2 &= a^2 \\ (x_e - a \cos \theta_3)^2 + (y_e - a \sin \theta_3)^2 &= a^2 \end{aligned} \quad (4)$$

These constraint equations can be simplified to

$$\begin{aligned} x_e &= 2a \cos \frac{\theta_1 - \theta_3}{2} \cos \frac{\theta_1 + \theta_3}{2} \\ y_e &= 2a \cos \frac{\theta_1 - \theta_3}{2} \sin \frac{\theta_1 + \theta_3}{2} \end{aligned} \quad (5)$$

Further simplifications, leads to:

$$\begin{Bmatrix} x_e \\ y_e \end{Bmatrix} = a \begin{bmatrix} \cos \theta_1 & \cos \theta_3 \\ \sin \theta_1 & \sin \theta_3 \end{bmatrix} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} \quad (6)$$

In order to determine the inverse kinematic equations of the device both the forms of the forward

kinematic equations developed earlier need to be utilized. For instance, from the expansion and subsequent simplification of equation (4), it can be deduced that

$$\theta_1 + \theta_3 = 2 \cot^{-1} \left(\frac{x_e}{y_e} \right) \quad (7)$$

In a similar manner, squaring and adding equation (5) yields

$$\theta_1 - \theta_3 = \cos^{-1} \left(\frac{x_e^2 + y_e^2 - 2a^2}{2a^2} \right) \quad (8)$$

From equations (7) and (8), it can then be deduced that

$$\begin{Bmatrix} \theta_1 \\ \theta_3 \end{Bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{Bmatrix} 2 \cot^{-1} \left(\frac{x_e}{y_e} \right) \\ \cos^{-1} \left(\frac{x_e^2 + y_e^2 - 2a^2}{2a^2} \right) \end{Bmatrix} \quad (9)$$

Once the active joint angles θ_1 and θ_3 are known, the passive joint angles θ_2 and θ_4 can be determined by direct inversion of the forward kinematic equations given in equation (3). In doing so, we get $\theta_2 = \cos^{-1} \left(\frac{x_e}{a} - \cos \theta_1 \right)$ and $\theta_4 = \cos^{-1} \left(\frac{x_e}{a} - \cos \theta_3 \right)$.

The Jacobian matrix, $\mathbf{J}(\Theta_{ms})$, that relates the vector of active joint velocities $\dot{\Theta}_{ms}$ to the vector of end-effector velocities $\dot{\mathbf{X}}$, and the vector of joint torques to vector of end-effector forces can be obtained by taking the time-derivative of the forward kinematic equations as given in equation (6). The vector of end-effector velocities is accordingly given by:

$$\underbrace{\begin{Bmatrix} \dot{x}_e \\ \dot{y}_e \end{Bmatrix}}_{\dot{\mathbf{X}}} = a \underbrace{\begin{bmatrix} -\sin \theta_1 & -\sin \theta_3 \\ \cos \theta_1 & \cos \theta_3 \end{bmatrix}}_{\mathbf{J}(\Theta_{ms})} \underbrace{\begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_3 \end{Bmatrix}}_{\dot{\Theta}_{ms}} \quad (10)$$

From this the Jacobian matrix, $\mathbf{J}(\Theta_{ms})$, of the device can be identified as

$$\mathbf{J}(\Theta_{ms}) = a \begin{bmatrix} -\sin \theta_1 & -\sin \theta_3 \\ \cos \theta_1 & \cos \theta_3 \end{bmatrix} \quad (11)$$

Dynamics formulation

The dynamics of parallel manipulators and likewise haptic devices can be expressed through various methods, including the Newton-Euler formulation, the principle of virtual work, and the Lagrangian formulation.⁵⁷ The Newton-Euler approach relies on the direct application of Newtons dynamic equations to each link of the manipulator, while the Lagrangian approach derives the dynamic equations based on the system's kinetic and potential energies. For devices with simple architecture and fewer DOF, the

Lagrangian dynamics formulation proves to be particularly advantageous. In view of this, the dynamics of the 2-DOF haptic device under consideration will be derived using the Lagrangian approach.

Once the generalized coordinates, Θ , have been selected, the initial step in the Lagrangian formulation involves determining the scalar function Lagrangian, $\mathcal{L}(\Theta, \dot{\Theta})$. The Lagrangian, $\mathcal{L}(\Theta, \dot{\Theta})$, is defined as the difference between the system's kinetic energy, $\mathcal{K}(\Theta, \dot{\Theta})$, and the potential energy, $\mathcal{P}(\Theta)$. For an n -DOF non-redundant device, the Lagrangian, $\mathcal{L}(\Theta, \dot{\Theta})$ is given as

$$\mathcal{L}(\Theta, \dot{\Theta}) = \sum_{i=1}^n (\mathcal{K}_i(\Theta, \dot{\Theta}) - \mathcal{P}_i(\Theta)) \quad (12)$$

In line with the Lagrangian formulation, the dynamics of the 2-DOF haptic device, with consideration to the generalized joint coordinates, Θ , can then be expressed as⁵⁸

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_i} \right) - \frac{\partial \mathcal{L}}{\partial \theta_i} = \tau_i \quad i = 1, 2, 3, 4 \quad (13)$$

where, τ_i , is the i^{th} joint torque.

As mentioned previously, the device maintains a horizontal position throughout the virtual wall interaction, resulting in the potential energy component, $\mathcal{P}(\Theta)$, being zero. Accordingly the Lagrangian, $\mathcal{L}(\Theta, \dot{\Theta})$, in this particular case, is equivalent to the total kinetic energy component, $\mathcal{K}(\Theta, \dot{\Theta})$, of the system.

Due to its symmetrical geometry, the kinetic energy, $\mathcal{K}(\Theta, \dot{\Theta})$, of the 2-DOF haptic device can be computed by treating it as a combination of two serial chains with the 2R architecture.⁵⁹ Thus, $\mathcal{K}(\Theta, \dot{\Theta})$ can be expressed as

$$\mathcal{K}(\Theta, \dot{\Theta}) = \mathcal{K}_{\{2R\}^1} + \mathcal{K}_{\{2R\}^2} \quad (14)$$

where, $\mathcal{K}_{\{2R\}^1}$ and $\mathcal{K}_{\{2R\}^2}$ represent the kinetic energy components of chain 1, which includes link 1 and link 2, and chain 2, which comprises of link 3 and link 4, respectively. For both the serial chains, the kinetic energy, $\mathcal{K}_{\{2R\}^i}$, can be evaluated by noting that the base link undergoes pure rotational motion, while the other link experiences a combination of rotational and translational motion. For the i^{th} link, denoting I_{bi} as the mass moment of inertia about the base, I_{cmi} as the mass moment of inertia about the center of mass, and $\dot{x}_{cmi}$ and $\dot{y}_{cmi}$ as the X and Y coordinates of the center of mass, the following expression for the kinetic energy of chain 1 can be written

$$\mathcal{K}_{\{2R\}^1} = \frac{1}{2} [I_{b1} \dot{\theta}_1^2 + I_{cm2} \dot{\theta}_2^2 + m_2 (\dot{x}_{cm2}^2 + \dot{y}_{cm2}^2)] \quad (15)$$

Assuming the links to be thin cylindrical rods with homogeneous composition, the above expression for $\mathcal{K}_{\{2R\}^1}$ yields

$$\mathcal{K}_{\{2R\}^1} = \frac{1}{2} \left[\left(\frac{m_1 a^2}{3} + m_2 a^2 \right) \dot{\theta}_1^2 + \left(\frac{m_2 a^2}{12} + \frac{m_2 a^2}{4} \right) \dot{\theta}_2^2 + m_2 a^2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \right] \quad (16)$$

Likewise, for the second chain, we obtain

$$\mathcal{K}_{\{2R\}^2} = \frac{1}{2} \left[\left(\frac{m_3 a^2}{3} + m_4 a^2 \right) \dot{\theta}_3^2 + \left(\frac{m_4 a^2}{12} + \frac{m_4 a^2}{4} \right) \dot{\theta}_4^2 + m_4 a^2 \dot{\theta}_3 \dot{\theta}_4 \cos(\theta_3 - \theta_4) \right] \quad (17)$$

With the expressions for the kinetic energy terms, $\mathcal{K}_{\{2R\}^1}$ and $\mathcal{K}_{\{2R\}^2}$ derived, we can now construct the Lagrangian, $\mathcal{L}(\Theta, \dot{\Theta})$, as:

$$\begin{aligned} \mathcal{L}(\Theta, \dot{\Theta}) = & \frac{1}{2} \left[\left(\frac{m_1 a^2}{3} + m_2 a^2 \right) \dot{\theta}_1^2 + \left(\frac{m_2 a^2}{12} + \frac{m_2 a^2}{4} \right) \dot{\theta}_2^2 + m_2 a^2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) + \left(\frac{m_3 a^2}{3} + m_4 a^2 \right) \dot{\theta}_3^2 \right. \\ & \left. + \left(\frac{m_4 a^2}{12} + \frac{m_4 a^2}{4} \right) \dot{\theta}_4^2 + m_4 a^2 \dot{\theta}_3 \dot{\theta}_4 \cos(\theta_3 - \theta_4) \right] \end{aligned} \quad (18)$$

By repeatedly applying equation (13) to equation (18) for various joint angles and velocities, and then consolidating the resulting equations, the dynamic equations of the 2-DOF device can be reduced to the following vector-matrix representation:

$$\mathbf{M}(\Theta) \ddot{\Theta} + \mathbf{C}(\Theta, \dot{\Theta}) = \tau \quad (19)$$

where, $\mathbf{M}(\Theta) \in \mathbb{R}^{4 \times 4}$ is the mass matrix, $\mathbf{C}(\Theta, \dot{\Theta}) \in \mathbb{R}^{4 \times 1}$ is the vector encompassing the coriolis and centrifugal terms, and $\tau \in \mathbb{R}^{4 \times 1}$ is the vector of joint torques corresponding to the generalized coordinates, Θ . The explicit expressions for $\mathbf{M}(\Theta)$ and $\mathbf{C}(\Theta, \dot{\Theta})$ are given as:

$$\mathbf{M}(\Theta) = \begin{bmatrix} \left(\frac{m_1}{3} + m_2 \right) a^2 & \frac{m_2}{2} a^2 \cos(\theta_1 - \theta_2) & 0 & 0 \\ \frac{m_2}{2} a^2 \cos(\theta_1 - \theta_2) & \left(\frac{m_2}{12} + \frac{m_2}{4} \right) a^2 & 0 & 0 \\ 0 & 0 & \left(\frac{m_3}{3} + m_4 \right) a^2 & \frac{m_4}{2} a^2 \cos(\theta_3 - \theta_4) \\ 0 & 0 & \frac{m_4}{2} a^2 \cos(\theta_3 - \theta_4) & \left(\frac{m_4}{12} + \frac{m_4}{4} \right) a^2 \end{bmatrix} \quad (20)$$

$$\mathbf{C}(\Theta, \dot{\Theta}) = \begin{bmatrix} \frac{m_2}{2} a^2 \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) \\ -\frac{m_2}{2} a^2 \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) \\ \frac{m_4}{2} a^2 \dot{\theta}_4^2 \sin(\theta_3 - \theta_4) \\ -\frac{m_4}{2} a^2 \dot{\theta}_3^2 \sin(\theta_3 - \theta_4) \end{bmatrix} \quad (21)$$

The dynamics equations for the 2-DOF device, as presented in equation (19), describe the device dynamics in terms of the generalized coordinates, Θ , that encompass both the active and passive joint angles. To express the device dynamics solely in terms of the active joint angles that is, the minimum set of generalized coordinates, Θ_{ms} , a transformation matrix relating $\dot{\Theta}$ and $\dot{\Theta}_{\text{ms}}$ must be derived.⁵⁹ Such a transformation matrix can be obtained by utilizing the forward kinematic equations of the device, which were previously developed in section 4.

Considering the following form of the forward kinematics equations (see equation (3))

$$\begin{Bmatrix} x_e \\ y_e \end{Bmatrix} = a \begin{bmatrix} \cos \theta_1 & \cos \theta_2 \\ \sin \theta_1 & \sin \theta_2 \end{bmatrix} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix} = a \begin{bmatrix} \cos \theta_3 & \cos \theta_4 \\ \sin \theta_3 & \sin \theta_4 \end{bmatrix} \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$$

The time derivative of the above equations is given as:

$$\begin{Bmatrix} \dot{x}_e \\ \dot{y}_e \end{Bmatrix} = a \begin{bmatrix} -\sin \theta_1 & -\sin \theta_2 \\ \cos \theta_1 & \cos \theta_2 \end{bmatrix} \begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{Bmatrix} = a \begin{bmatrix} -\sin \theta_3 & -\sin \theta_4 \\ \cos \theta_3 & \cos \theta_4 \end{bmatrix} \begin{Bmatrix} \dot{\theta}_3 \\ \dot{\theta}_4 \end{Bmatrix}$$

The successive elimination of $\dot{\theta}_2$ and $\dot{\theta}_4$ from the above equations yield

$$\begin{Bmatrix} \dot{\theta}_2 \\ \dot{\theta}_4 \end{Bmatrix} = \begin{bmatrix} -\frac{\sin(\theta_1 - \theta_4)}{\sin(\theta_2 - \theta_4)} & \frac{\sin(\theta_3 - \theta_4)}{\sin(\theta_2 - \theta_4)} \\ \frac{\sin(\theta_1 - \theta_2)}{\sin(\theta_4 - \theta_2)} & -\frac{\sin(\theta_3 - \theta_2)}{\sin(\theta_4 - \theta_2)} \end{bmatrix} \begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_3 \end{Bmatrix} \quad (22)$$

Thus, the transformation relating the generalized joint velocities, $\dot{\Theta}$, and the minimum set of joint velocities, $\dot{\Theta}_{ms}$, can be expressed as follows:

$$\begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \end{Bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ -\frac{\sin(\theta_1 - \theta_4)}{\sin(\theta_2 - \theta_4)} & \frac{\sin(\theta_3 - \theta_4)}{\sin(\theta_2 - \theta_4)} \\ 0 & 1 \\ \frac{\sin(\theta_1 - \theta_2)}{\sin(\theta_4 - \theta_2)} & -\frac{\sin(\theta_3 - \theta_2)}{\sin(\theta_4 - \theta_2)} \end{bmatrix}}_{\Psi} \begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_3 \end{Bmatrix} \quad (23)$$

More compactly,

$$\dot{\Theta} = \Psi \dot{\Theta}_{ms} \quad (24)$$

where, $\Psi \in \mathbb{R}^{4 \times 2}$, is the transformation matrix that relates $\dot{\Theta}$ with $\dot{\Theta}_{ms}$ and is defined as the pre-multiplying matrix in equation (23).

With the application of this transformation, the dynamics equations presented earlier in equation (19) are reformulated into the following form⁵⁹:

$$\underbrace{\Psi^T M \Psi}_{M_{ms}} \ddot{\Theta}_{ms} + \underbrace{\Psi^T (C + M \dot{\Psi} \dot{\Theta}_{ms})}_{C_{ms}} = \underbrace{\Psi^T \tau}_{\tau_{ms}} \quad (25)$$

This can be written simply as

$$M_{ms} \ddot{\Theta}_{ms} + C_{ms} = \tau_{ms} \quad (26)$$

where, $M_{ms} \in \mathbb{R}^{2 \times 2}$ is the mass matrix, $C_{ms} \in \mathbb{R}^{2 \times 1}$ is the vector accounting for the nonlinear inertia terms, and $\tau_{ms} \in \mathbb{R}^{2 \times 1}$ is the vector of active joint torques, all corresponding to the minimum set of generalized coordinates, Θ_{ms} . It is essential to highlight that the components of the vector, τ_{ms} , that is, the active joint torques, emerge from a combination of various physical sources, including the human operator, device dissipation, link compliance, etc., and the forces arising from interactions with the virtual environment.

Decoupling of the dynamics equations

The dynamics equations of the 2-DOF haptic device formulated in the preceding section (see equation (26)) exhibit coupling between the admissible DOFs, namely, the X and Y DOFs, in the operational space. This dynamic coupling becomes evident when observing the non-zero values of the off-diagonal terms of

the mass matrix, M_{ms} , which appears in the dynamic equations as described by equation (26). For a generic multibody system, a detailed interpretation of the various elements of the mass matrix can be found in Hirschhorn and Kővecses.⁶⁰ This dynamic coupling is a consequence of the multi-DOF nature and parallel architecture of the 2-DOF device.^{10,60} In general, for multi-DOF parallel devices, the dynamic coupling implies that the motion and forces imparted in one operational space direction can influence and impact the behavior in all the admissible directions.⁵⁹ Thus, in scenarios wherein the 2-DOF haptic device renders a virtual wall along either of the operational space direction, X or Y , the existence of dynamic coupling prohibits analyzing the devices stability in that isolated direction. This necessitates a reformulation of the dynamics equations in a manner that decouples the dynamics of the rendered and admissible directions.

In this section, we employ the dynamic decoupling strategy proposed in Shayan-Amin et al.¹⁰ and Shayan-Amin⁵⁹ for the generic class of multi-DOF haptic devices. The approach involves the decomposition of the dynamics equations into a set of two distinct equations: one corresponding to the dynamics of physically rendered directions and the other describing the dynamics of the remaining admissible directions in the operational space. The key concept behind this approach is to find a transformation that ensures that the mass matrix associated with the dynamic equations assumes a diagonal form. In Shayan-Amin et al.¹⁰ and Shayan-Amin⁵⁹, the authors have validated their approach using a 2-DOF parallel haptic device. However, it differs from the one considered here in two aspects: our device has a base link length of zero, and the other link lengths are uniform, whereas the one in Shayan-Amin et al.¹⁰ and Shayan-Amin⁵⁹ has a non-zero base length and non-uniform link dimensions. Additionally, the authors considered a uniform-rate control strategy, while our focus here is on the dual-rate control strategy.

The process of decoupling the dynamics equations becomes more straightforward when considering the linearized dynamics equations of the device. The dynamics equations obtained previously in equation (26) can be linearized about an equilibrium configuration near the virtual wall, Θ_{lms} , when small device motions about the virtual wall are involved. In reality, such an equilibrium configuration corresponds to the rendering scenarios wherein the net physical forces, such as those from the human operator, physical dissipation, etc., acting on the device, are nearly balanced by the forces stemming from the virtual wall interaction. In accordance with Shayan-Amin,⁵⁹ such a situation is referred to as the condition of *haptic equilibrium*.

The linearized form of the dynamics equations given by equation (26) can be written as

$$\mathbf{M}_{\text{lms}} \ddot{\boldsymbol{\Theta}}_{\text{lms}} = \boldsymbol{\tau}_{\text{lms}} \quad (27)$$

where, the mass matrix, $\mathbf{M}_{\text{lms}} \in \mathbb{R}^{2 \times 2}$, and the vector of active joint torques, $\boldsymbol{\tau}_{\text{lms}} \in \mathbb{R}^{2 \times 1}$, correspond to the equilibrium configuration, $\boldsymbol{\Theta}_{\text{lms}} \in \mathbb{R}^{2 \times 1}$, of the haptic device. The linearized mass matrix $\mathbf{M}_{\text{lms}}$ is equal to the mass matrix $\mathbf{M}_{\text{ms}}$ but evaluated at the equilibrium configuration, that is, $\mathbf{M}_{\text{lms}} = \mathbf{M}_{\text{ms}}(\boldsymbol{\Theta}_{\text{lms}})$.

As haptic devices usually render virtual walls in the operational space instead of the joint space, it is crucial to express the device dynamics in terms of the operational space coordinates. Utilizing the equivalence between the device's kinetic energy in the joint space and the operational space coordinates, we have the following expression⁵⁷

$$\frac{1}{2} \dot{\boldsymbol{\Theta}}_{\text{lms}}^T \mathbf{M}_{\text{lms}} \dot{\boldsymbol{\Theta}}_{\text{lms}} = \frac{1}{2} \dot{\mathbf{X}}^T \boldsymbol{\Lambda} \dot{\mathbf{X}} \quad (28)$$

where, $\boldsymbol{\Lambda} \in \mathbb{R}^{2 \times 2}$ is used to represent the mass matrix in the operational space coordinates. Recalling that the Jacobian Matrix, $\mathbf{J}$, as defined in equation (11), relates the vector of joint torques to the vector of end-effector forces, we can express equation (27) as follows:

$$\underbrace{\mathbf{J}^{-T} \mathbf{M}_{\text{lms}} \mathbf{J}^{-1}}_{\boldsymbol{\Lambda}} \ddot{\mathbf{X}} = \underbrace{\mathbf{J}^{-T} (\boldsymbol{\tau}_{\text{phy}} + \boldsymbol{\tau}_{\text{vir}})}_{\mathbf{F}_{\text{phy}} + \mathbf{F}_{\text{vir}}} \quad (29)$$

where, $\boldsymbol{\tau}_{\text{phy}}$ and $\boldsymbol{\tau}_{\text{vir}}$ represent the active joint torques, while $\mathbf{F}_{\text{phy}}$ and $\mathbf{F}_{\text{vir}}$ denote the forces acting on the end-effector, arising from physical sources and virtual interaction, respectively.

Consequently, the dynamics of the device in terms of the operational space formulation are expressed as follows:

$$\boldsymbol{\Lambda} \ddot{\mathbf{X}} = \mathbf{F}_{\text{phy}} + \mathbf{F}_{\text{vir}} \quad (30)$$

Further, these linearized dynamics equations can be expressed as

$$\begin{bmatrix} \boldsymbol{\Lambda}_{xx} & \boldsymbol{\Lambda}_{xy} \\ \boldsymbol{\Lambda}_{yx} & \boldsymbol{\Lambda}_{yy} \end{bmatrix} \begin{Bmatrix} \ddot{x}_e \\ \ddot{y}_e \end{Bmatrix} = \mathbf{F}_{\text{phy}} + \mathbf{F}_{\text{vir}} \quad (31)$$

Decomposing the Jacobian matrix, $\mathbf{J}$, in the form

$$\mathbf{J} = \begin{Bmatrix} \mathbf{J}_X \\ \mathbf{J}_Y \end{Bmatrix} \quad (32)$$

We can write

$$\boldsymbol{\Lambda}^{-1} = \mathbf{J} \mathbf{M}_{\text{lms}}^{-1} \mathbf{J}^T = \begin{bmatrix} \mathbf{J}_X \mathbf{M}_{\text{lms}}^{-1} \mathbf{J}_X^T & \mathbf{J}_X \mathbf{M}_{\text{lms}}^{-1} \mathbf{J}_Y^T \\ \mathbf{J}_Y \mathbf{M}_{\text{lms}}^{-1} \mathbf{J}_X^T & \mathbf{J}_Y \mathbf{M}_{\text{lms}}^{-1} \mathbf{J}_Y^T \end{bmatrix} \quad (33)$$

In the above equation, if $\mathbf{J}_Y$ is chosen such that $\mathbf{J}_X \mathbf{M}_{\text{lms}}^{-1} \mathbf{J}_Y^T = 0$, then the off-diagonal terms in the mass matrix, $\boldsymbol{\Lambda}$, would vanish and consequently the dynamic coupling between the X and Y DOFs would be eliminated.^{10,59} Moreover, if the 2-DOF device renders a virtual wall along the Y -direction alone then the dynamics equations can be decomposed into the following two parts:

$$\boldsymbol{\Lambda}_{xx} \ddot{x}_e = F_{\text{phy}} \quad (34)$$

$$\boldsymbol{\Lambda}_{yy} \ddot{y}_e = F_{\text{phy}} + F_{\text{vir}} \quad (35)$$

where, for the decoupled configuration, $\boldsymbol{\Lambda}_{xx} = (\mathbf{J}_X \mathbf{M}_{\text{lms}}^{-1} \mathbf{J}_X^T)^{-1}$ and $\boldsymbol{\Lambda}_{yy} = (\mathbf{J}_Y \mathbf{M}_{\text{lms}}^{-1} \mathbf{J}_Y^T)^{-1}$.

In the above equations, it is to be noted that the virtual forces, F_{vir} , need to be included only in the dynamics equations of the operational space coordinate (in this case the Y -coordinate) along which the virtual wall is rendered and not for the other admissible direction.

Exact discrete-time stability analysis

In this section, we investigate the stability of the 2-DOF haptic device, assuming that its motion is confined solely to the Y -direction, while the X -coordinate remains fixed at zero throughout. Accordingly, the Y -direction corresponds to the rendered direction, while the X -direction represents the admissible direction. Furthermore, in this configuration, we assume that the device renders a virtual wall, modeled as a parallel spring-damper system with stiffness K_y and damping B_y . Therefore, for this specific scenario, we solely need to examine the decoupled dynamics equations associated with the Y -coordinate. From equation (35), the dynamics equation for such a case are given as

$$\boldsymbol{\Lambda}_{yy} \ddot{y}_e = F_{\text{phy}} + F_{\text{vir}} \quad (36)$$

As mentioned earlier, the physical forces, F_{phy} , acting on the haptic device stem from the human operator input, physical dissipation and compliance of the device. It is however known that instabilities in haptic devices tend to occur predominantly at frequencies that lie beyond the bandwidth of human voluntary movement and the involuntary tremor frequencies.^{21,61} More importantly, previous studies indicate that a human operator can potentially enhance the stability of the haptic device by reinforcing their hand impedance onto the device impedance.^{31,62} In the ensuing analysis, we will therefore consider a worst-case scenario wherein we assume that the human operator holds the device with minimal or negligible grasping force, thus focusing on the *uncoupled stability*.⁴⁰ Moreover, due to the rigid nature of the device links and negligible transmission flexibility, the forces

resulting from device compliance are also negligible. Consequently, the primary source of physical force acting on the device arises from joint dissipation, which we model as viscous damping.

Accordingly, at any given time instant, t , the dynamics of the device while in sustained contact with the virtual wall can be represented by the following equation

$$\Lambda_{yy}\ddot{y}_e(t) + b_{yy}\dot{y}_e(t) + F_K(t - \lambda) + F_B(t - \lambda) = 0 \quad (37)$$

where, b_{yy} is the viscous damping as experienced at the end-effector while moving in the Y -direction, F_K and F_B are the virtual interaction forces due to the virtual stiffness and virtual damping, respectively, and λ is the time duration by which the feedback forces are delayed.

As the virtual wall is implemented at discrete sampling instants, the feedback forces, $F_K(t)$ and $F_B(t)$, are governed by:

$$F_K(t) = F_K(jT_K) = K_y y_e[jT_K] \quad \forall \quad t \in [jT_K, (j+1)T_K] \quad (38)$$

$$F_B(t) = F_B(jT_B) = B_y \dot{y}_e[jT_B] \quad \forall \quad t \in [jT_B, (j+1)T_B] \quad (39)$$

where, in the above equations a zero-order hold (ZOH) is considered for the digital to analog conversion which is the most common implementation in haptics applications.^{9,17,31}

In order to analyze the stability of the system we make use of the state-space framework and for that we choose the end-effector position and velocity as the two state variables, that is, we define:

$$y_1 := y_e \quad \text{and} \quad y_2 := \dot{y}_e \quad (40)$$

With this choice, the device dynamics as defined by the differential equation in equation (37), can be transformed into the following continuous-time state-space representation

$$\begin{cases} \dot{\mathbf{Y}}(t) = \mathbf{A}\mathbf{Y}(t) + \mathbf{B}U(t - \lambda) \\ Z(t) = \mathbf{C}\mathbf{Y}(t) \end{cases} \quad (41)$$

where, $\mathbf{Y} \in \mathbb{R}^{2 \times 1}$, $U \in \mathbb{R}$, and $Z \in \mathbb{R}$ are the state vector, the control input, and the system output, respectively. Furthermore, $\mathbf{A} \in \mathbb{R}^{2 \times 2}$, $\mathbf{B} \in \mathbb{R}^{2 \times 1}$, and $\mathbf{C} \in \mathbb{R}^{1 \times 2}$ represent the state matrix, the control vector, and the output vector, respectively. For the system under consideration, they are given by:

$$\begin{aligned} \mathbf{Y} &= \{y_1 \ y_2\}^T; \quad \mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{b_{yy}}{\Lambda_{yy}} \end{bmatrix}; \\ \mathbf{B} &= \left\{ 0 \quad \frac{1}{\Lambda_{yy}} \right\}^T \quad \text{and} \quad \mathbf{C} = \{1 \ 0\} \end{aligned} \quad (42)$$

Owing to the presence of two sampling rates in the dual-rate haptics controller, the initial step in analyzing its stability involves discretization of the continuous-time state-space equations, as described by equation (41), in accordance with the position loop's sampling period, T_K . Subsequently, the inner feedback loop, which is the position loop, needs to be closed to account for the virtual stiffness. Following this, the resulting system needs to be rediscritized with a sampling period of T_B in order to conform with the sampling rate of the outer feedback loop, that is, the velocity loop. The final step involves the closing of the velocity loop to account for the virtual damping. Additionally, the presence of time delay in the position and velocity feedback loops requires proper augmentation of the resulting state vectors. The detailed procedure involved in this analysis can be found in our previous work which dealt with the stability analysis of single-DOF systems.⁴⁸ In this section, we will only highlight the main results.

Following the ZOH assumption, the discretization of the continuous-time state-space equations, as given by equation (41), with respect to T_K , results in the following discrete-time state-space equations:

$$\begin{cases} \mathbf{Y}[(j+1)T_K] = \Phi \mathbf{Y}[jT_K] + \Gamma(-F_K[(j-\lambda_K)T_K] \\ \quad \quad \quad + \Gamma(-F_B[(j-\lambda_B)T_K]) \\ Z[jT_K] = \Omega \mathbf{Y}[jT_K] \end{cases} \quad (43)$$

The matrices Φ , Γ , and Ω involved in the discretized equations, equation (43), are defined as:

$$\Phi = e^{\mathbf{A}T_K}; \quad \Gamma = \left(\int_0^{T_K} e^{\mathbf{A}\eta} d\eta \right) \mathbf{B} \quad \text{and} \quad \Omega = \mathbf{C} \quad (44)$$

With their explicit expressions given by^{48,63}:

$$\begin{aligned} \Phi &= \begin{bmatrix} 1 & \frac{\Lambda_{yy} \left(1 - e^{-\frac{b_{yy}T_K}{\Lambda_{yy}}} \right)}{b_{yy}} \\ 0 & e^{-\frac{b_{yy}T_K}{\Lambda_{yy}}} \end{bmatrix} \quad \text{and} \\ \Gamma &= \begin{bmatrix} \frac{T_K}{b_{yy}} + \frac{\Lambda_{yy}}{b_{yy}^2} \left(e^{-\frac{b_{yy}T_K}{\Lambda_{yy}}} - 1 \right) \\ \frac{1}{b_{yy}} \left(1 - e^{-\frac{b_{yy}T_K}{\Lambda_{yy}}} \right) \end{bmatrix} \end{aligned} \quad (45)$$

In accordance with the governing equation, equation (38), for the feedback force, F_K , and by introducing λ_K number of new state variables to account for the time delay, $\lambda = \lambda_K T_K$, in the position loop, the discretized state-space equations given by equation (43), can be expressed in augmented form as:

$$\begin{cases} \mathbf{Y}_{\text{aug}}[(j+1)T_K] = \mathbf{\Phi}_{\text{aug}} \mathbf{Y}_{\text{aug}}[jT_K] + \mathbf{\Gamma}_{\text{aug}} F_B[(j-\lambda_B)T_K] \\ Z[jT_K] = \mathbf{\Omega}_{\text{aug}} \mathbf{Y}_{\text{aug}}[jT_K] \end{cases} \quad (46)$$

where, the augmented matrices involved in the above equation are defined as

$$\begin{aligned} \mathbf{Y}_{\text{aug}}[jT_K] &= \begin{Bmatrix} \mathbf{Y}[jT_K] \\ y_3[jT_K] \\ y_4[jT_K] \\ \vdots \\ y_{\lambda_K+1}[jT_K] \\ y_{\lambda_K+2}[jT_K] \end{Bmatrix}; \\ \mathbf{\Phi}_{\text{aug}} &= \begin{bmatrix} \mathbf{\Phi} & -\mathbf{\Gamma}K_y\mathbf{\Omega} & \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & 1 & 0 & \dots & 0 \\ \mathbf{0} & \mathbf{0} & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ \mathbf{0} & \mathbf{0} & 0 & 0 & \dots & 1 \\ \mathbf{1} & \mathbf{0} & 0 & 0 & \dots & 0 \end{bmatrix}; \\ \mathbf{\Gamma}_{\text{aug}} &= \begin{Bmatrix} \mathbf{\Gamma} \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{Bmatrix} \quad \mathbf{\Omega}_{\text{aug}} = \begin{Bmatrix} \mathbf{\Omega} \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{Bmatrix}^T \end{aligned} \quad (47)$$

The augmented discrete-time state-space equation, as defined in equation (47), characterizes the combined dynamics of the haptic device and the position loop. This is of significance in evaluating the stability limits when only virtual spring is rendered and thereby when the virtual damping coefficient, and consequently the associated damping force, are both zero.

The existence of distinct sampling rates in the position and velocity loops of the controller necessitate that equation (47) be rediscritized that is, resampled from sampling period T_K to T_B before the velocity loop can be closed. As T_K and T_B have been assumed to be integrally related, the rediscritized state-space equations can be expressed as:

$$\begin{cases} \mathbf{Y}_{\text{aug}}[(j+1)T_B] = \mathbf{\Phi}_{\text{aug}}^r \mathbf{Y}_{\text{aug}}[jT_B] + \mathbf{\Gamma}_{\text{aug}}^r F_B[(j-\lambda_B)T_B] \\ Z[jT_B] = \mathbf{\Omega}_{\text{aug}} \mathbf{Y}_{\text{aug}}[jT_B] \end{cases} \quad (48)$$

where, the rediscritized matrices, $\mathbf{\Phi}_{\text{aug}}^r$ and $\mathbf{\Gamma}_{\text{aug}}^r$, involved in the above equation are defined as^{47,48}

$$\mathbf{\Phi}_{\text{aug}}^r = [\mathbf{\Phi}_{\text{aug}}]^N \quad \text{and} \quad \mathbf{\Gamma}_{\text{aug}}^r = \sum_{i=0}^{N-1} [\mathbf{\Phi}_{\text{aug}}]^i \mathbf{\Gamma}_{\text{aug}} \quad (49)$$

With the state-space equations now represented in terms of a single sample period T_B , it is possible to close the velocity feedback loop by utilizing the governing equation for the feedback force F_B , as defined in equation (39). Additionally, we introduce new state variables to take care of the time delay, λ_B , in the velocity loop. The resulting state-space equations that represent the dynamics of the overall dual-rate control loop can then be expressed as:

$$\begin{cases} \mathbf{Y}_{\text{cl}}[(j+1)T_B] = \mathbf{\Phi}_{\text{cl}} \mathbf{Y}_{\text{cl}}[jT_B] \\ Z[jT_B] = \mathbf{\Omega}_{\text{cl}} \mathbf{Y}_{\text{cl}}[jT_B] \end{cases} \quad (50)$$

These closed-loop state-space equations of the combined device-virtual environment system can now be used to assess the stability of the device for various values of virtual wall parameters. However, in order to determine the stability boundaries of the device it is imperative to first determine the value of the viscous damping coefficient b_{yy} that the end-effector experiences in the operational space.

Considering b_i as the viscous damping coefficient at the devices i^{th} joint, the viscous torque τ_{bi} at each joint can be expressed as⁶⁴

$$\begin{Bmatrix} \tau_{b1} \\ \tau_{b2} \\ \tau_{b3} \\ \tau_{b4} \end{Bmatrix} = \begin{bmatrix} b_1 & 0 & 0 & 0 \\ 0 & b_2 & 0 & 0 \\ 0 & 0 & b_3 & 0 \\ 0 & 0 & 0 & b_4 \end{bmatrix} \begin{Bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \end{Bmatrix} \quad (51)$$

More compactly, we can write

$$\tau_b = \mathbf{B}_J \dot{\Theta} \quad (52)$$

where, $\tau_b \in \mathbb{R}^{4 \times 1}$ and $\mathbf{B}_J \in \mathbb{R}^{4 \times 4}$ are the vector of joint viscous torques and the diagonal joint viscous damping matrix, respectively. It is to be noted, however, that both τ_b and $\mathbf{B}_J$ correspond to the generalized coordinates Θ .

The mapping between $\dot{\Theta}$ and $\dot{\Theta}_{\text{ms}}$ that was developed earlier in equation (24) can be used to express equation (52) in terms of the minimum set of joint velocities $\dot{\Theta}_{\text{ms}}$ as follows

$$\Psi^T \tau_b = \Psi^T \mathbf{B}_J \Psi \dot{\Theta}_{\text{ms}} \quad (53)$$

This can be simply written as

$$\tau_{\text{bms}} = \mathbf{B}_{\text{Jms}} \dot{\Theta}_{\text{ms}} \quad (54)$$

The transposed Jacobian matrix, $\mathbf{J}^T$, can be used to map the vector of joint viscous torques τ_{bms} to the

vector of viscous forces $\mathbf{F}_b$ experienced at the end-effector, as follows

$$\mathbf{F}_b = \mathbf{J}^{-T} \tau_{bms} = \mathbf{J}^{-T} \mathbf{B}_{Jms} \mathbf{J}^{-1} \dot{\mathbf{X}} \quad (55)$$

Using the notation, $\mathbf{B}_{xy} = \mathbf{J}^{-T} \mathbf{B}_{Jms} \mathbf{J}^{-1}$, we can write

$$\mathbf{F}_b = \mathbf{B}_{xy} \dot{\mathbf{X}} \quad (56)$$

In expanded form, this can be expressed as

$$\begin{Bmatrix} F_{bxx} \\ F_{byy} \end{Bmatrix} = \begin{bmatrix} b_{xx} & b_{xy} \\ b_{yx} & b_{yy} \end{bmatrix} \begin{Bmatrix} \dot{x}_e \\ \dot{y}_e \end{Bmatrix} \quad (57)$$

For decoupled motion in the two operational space directions, the off-diagonal elements of the viscous damping matrix $\mathbf{B}_{xy}$ must be zero.⁶⁴ Accordingly, the term b_{yy} appearing in $\mathbf{B}_{xy}$ gives the effective viscous damping experienced at the end-effector when the 2-DOF device is constrained to move along the Y -direction only. It is to be noted that the order of the overall state-space equations in equation (50) is $\lambda_K + \lambda_B + 3$, which becomes quite high for higher sampling frequencies and larger time delays, thereby making the computation of stability limits very demanding. To simplify the determination of stability boundaries, we employ a stability-preserving MOR technique in the next section.

Balanced truncation based MOR

The stability analysis of controllers with time-delayed feedback is complicated due to the high dimensionality of the system models involved. In multi-rate controllers, this complexity is further exacerbated when time-delay terms are part of distinctly sampled loops, necessitating independent consideration of each loop delay. This complexity was noted in the previous section while examining the stability of the 2-DOF dual-rate haptics controller using the exact discrete-time state-space framework. The issue primarily stems from the time delay terms in the two feedback loops and the corresponding state augmentation, unrelated to the actual device dynamics or virtual environment representation. It is important to note, however, that the augmented states are used to formulate the state-space equations in standard forms and do not represent any physical states. In our prior work,⁴⁸ on the stability analysis of 1-DOF time-delayed dual-rate haptics controllers, we demonstrated that the balanced truncation based MOR technique^{65,66} can be utilized to develop lower-order models approximating the behavior of the original higher-order models while preserving their stability. In this section, we extend the same MOR method for simplifying the stability analysis of the 2-DOF dual-rate controllers.

For the overall state-space equations of the 2-DOF dual-rate haptics controller of the form:

$$\begin{cases} \mathbf{Y}_{cl}[(j+1)T_B] = \Phi_{cl} \mathbf{Y}_{cl}[jT_B] + \Xi_{cl} F_B[jT_B] \\ Z[jT_B] = \Omega_{cl} \mathbf{Y}_{cl}[jT_B] \end{cases} \quad (58)$$

The discrete controllability, ω_c , and observability gramian, ω_o , are crucial for the balanced truncation method. For the state-space representation of the 2-DOF dual-rate haptics controllers in equation (58), they are given by⁶³:

$$\begin{aligned} \omega_c &= \sum_{j=0}^{\infty} \Phi_{cl}^j \Xi_{cl} \Xi_{cl}^T (\Phi_{cl}^T)^j \\ \omega_o &= \sum_{j=0}^{\infty} (\Phi_{cl}^T)^j \Omega_{cl}^T \Omega_{cl} \Phi_{cl}^j \end{aligned} \quad (59)$$

However, they are usually obtained by solving the discrete-time Lyapunov equations⁶³:

$$\begin{aligned} \Phi_{cl} \omega_c \Phi_{cl}^T - \omega_c + \Xi_{cl} \Xi_{cl}^T &= 0 \\ \Phi_{cl}^T \omega_o \Phi_{cl} - \omega_o + \Omega_{cl}^T \Omega_{cl} &= 0 \end{aligned} \quad (60)$$

The balanced truncation method relies on the balanced realization of the system – a basis dependent representation where ω_c and ω_o are diagonal and equal, that is,

$$\omega_c = \omega_o = \omega = \text{diag}\{\sigma_1, \sigma_2, \sigma_3, \sigma_{\lambda_K + \lambda_B + 3}\} \quad (61)$$

The physical significance of such a system realization is that each system state is equally controllable and observable simultaneously.⁶⁷ The square roots of the σ 's are known as the Hankel Singular Values (HSVs) of the system, representing the energy of the corresponding system state. A higher value signifies that the particular state dominates the system dynamics. Once the balanced realization of the system is obtained, the balanced truncation method retains system states with high HSVs while discarding states with lower HSVs. A detailed methodology for obtaining the balanced realization and hence the HSVs is provided in our earlier work.⁴⁸ The procedure involves using the Cholesky and Singular Value Decomposition to determine a similarity transformation that transforms the higher-order state-space representation to a balanced realization.^{66,68} Then the diagonalized gramian, ω , is partitioned such that it becomes block diagonal, with states having higher HSVs contained in the upper diagonal block. If the gramian is partitioned as:

$$\omega = \begin{bmatrix} \omega_{11} & \mathbf{0} \\ \mathbf{0} & \omega_{22} \end{bmatrix} \quad (62)$$

where, the upper block $\mathbf{\omega}_{11} = \text{diag}\{\sigma_1, \sigma_2, \sigma_p\}$ comprises of p states with dominant HSVs retained in the reduced model, while the lower block $\mathbf{\omega}_{22} = \text{diag}\{\sigma_{p+1}, \sigma_{p+2}, \sigma_{\lambda_K + \lambda_B + 3}\}$ contains the remaining non-dominant states that are discarded. Similarly, the system matrices are partitioned such that dominant and non-dominant states are separated:

$$\mathbf{\Phi}_{cl} = \begin{bmatrix} \mathbf{\Phi}_{cl,11} & \mathbf{\Phi}_{cl,12} \\ \mathbf{\Phi}_{cl,21} & \mathbf{\Phi}_{cl,22} \end{bmatrix}; \mathbf{\Xi}_{cl} = \{\mathbf{\Xi}_{cl1} \quad \mathbf{\Xi}_{cl2}\}^T, \text{ \& \ } \{\mathbf{\Omega}_{cl1} \quad \mathbf{\Omega}_{cl2}\} \quad (63)$$

Based on this partitioning, the p^{th} order reduced system model of the original $(\lambda_B + \lambda_K + 3)^{th}$ order model (see equation (58)) is given as⁶⁵:

$$\begin{cases} \mathbf{Y}_{cl,r}[(j+1)T_B] = \mathbf{\Phi}_{cl,11}\mathbf{Y}_{cl,r}[jT_B] + \mathbf{\Xi}_{cl1}\mathbf{F}_B[jT_B] \\ \mathbf{Z}_r[jT_B] = \mathbf{\Omega}_{cl1}\mathbf{Y}_{cl,r}[jT_B] \end{cases} \quad (64)$$

It is important to note that although this p^{th} order reduced model is an approximation to the original higher order model in equation (58), both the models exhibit the same stability limits.

Results and discussions

In order to assess the stability limits of the 2-DOF haptic device, we employed the discrete-time variant of the Lyapunov theory. In accordance with this theory, the coupled device-environment dynamics, as described in equations (50) and (64), exhibits stable behavior under specific combinations of virtual wall parameters and sampling rates if there exists a single positive definite matrix $\mathbf{E}$, corresponding to a specified positive definite matrix $\mathbf{F}$, while satisfying the following discrete-time equation.⁶³

$$\underbrace{\mathbf{\Phi}_{cl}\mathbf{E}\mathbf{\Phi}_{cl}^T - \mathbf{E} + \mathbf{F}}_{(\lambda_K + \lambda_B + 3)^{th} \text{ order}} = \mathbf{0} \quad \text{high-order model}$$

$$\underbrace{\mathbf{\Phi}_{cl,11}\mathbf{E}\mathbf{\Phi}_{cl,11}^T - \mathbf{E} + \mathbf{F}}_{p^{th} \text{ order}} = \mathbf{0} \quad \text{low-order model} \quad (65)$$

Following equation (65), we evaluated the stability boundaries of the 2-DOF device by incrementally changing the values of the virtual damping coefficient B_y , and for each selected value of B_y , identifying the maximum value of the virtual stiffness coefficient K_y , for which the condition yielded a positive definite matrix $\mathbf{E}$. In all cases, we assumed that the matrix $\mathbf{F}$ was equal to an identity matrix of same size. In order to present the results, we made use of the non-dimensional parameters recommended in Gil et al.,²⁰

Table 1. Non-dimensional parameters in accordance with Gil et al.,²⁰ and Mashayekhi et al.²³

Parameter	Variable	Dimensionless variable
Delay in position loop	λ	$\lambda_K := \frac{\lambda}{T_K}$
Delay in velocity loop	λ	$\lambda_B := \frac{\lambda}{T_B}$
Physical damping	b_{yy}	$\delta := \frac{b_{yy}T_B}{\lambda_{yy}}$
Virtual damping	B_y	$\beta := \frac{B_yT_B}{\lambda_{yy}}$
Virtual stiffness	K_y	$\alpha := \frac{K_yT_B^2}{\lambda_{yy}}$

and Mashayekhi et al.²³ which are reproduced here in Table 1.

The stability boundaries of the 2-DOF haptic device under the dual-rate sampling strategy while rendering a single virtual wall along the Y-direction are shown in Figure 3 for a variety of sampling rate combinations and time delays. The virtual wall location is chosen to have the coordinates $\mathbf{X} = [0.145]^T$. The boundaries have been obtained by considering the physical parameters of the 2-DOF haptic device reported in Koul et al.¹⁴ and Koul²⁷ with $a = 10$ cm, $m'_s = 0.035$ kg, and $b'_s = b = 0.00123$ Nms/rad. In consistency with the results obtained in previous works for single-DOF devices^{20,23,31,33,48} the stability boundaries obtained here for the 2-DOF counterpart also exhibit similar trends. Notably, the stable virtual stiffness K_y (non-dimensional α) increases as the sampling rate f_K of the position loop increases whilst maintaining a fixed velocity loop sampling rate f_B . Furthermore, for fixed controller sampling rates f_K and f_B , an increase in time delay λ reduces both the stability margins of the controller. It is important to mention that these stability boundaries have been obtained using both the higher-order and lower-order system models. As was the case in Ganiny et al.,⁴⁸ the application of the MOR technique to the 2-DOF device once again reveals that the stability boundaries can be determined based on reduced third-order models ($p = 3$), regardless of the magnitude of time delay in the controller or the ratio of the sampling rates employed.

To validate the outcomes of our theoretical findings, we compared our theoretical stability limits with those obtained in Koul.²⁷ The experimental results, as reported in Koul,²⁷ were acquired by utilizing a low-pass Butterworth filter in the velocity loop, resulting in a time delay of approximately 1 ms. Accordingly, for the sake of comparison, we considered a time delay of 2 sample periods in the velocity loop, while the position loop was assumed to be undelayed. The results of this comparison are depicted in Figure 4 for a variety of sample rate combinations. As can be seen the theoretical analysis is able to predict the stability limits to a relatively good degree of accuracy, albeit with a slight conservatism compared to the experimental results. This conservatism in the theoretical

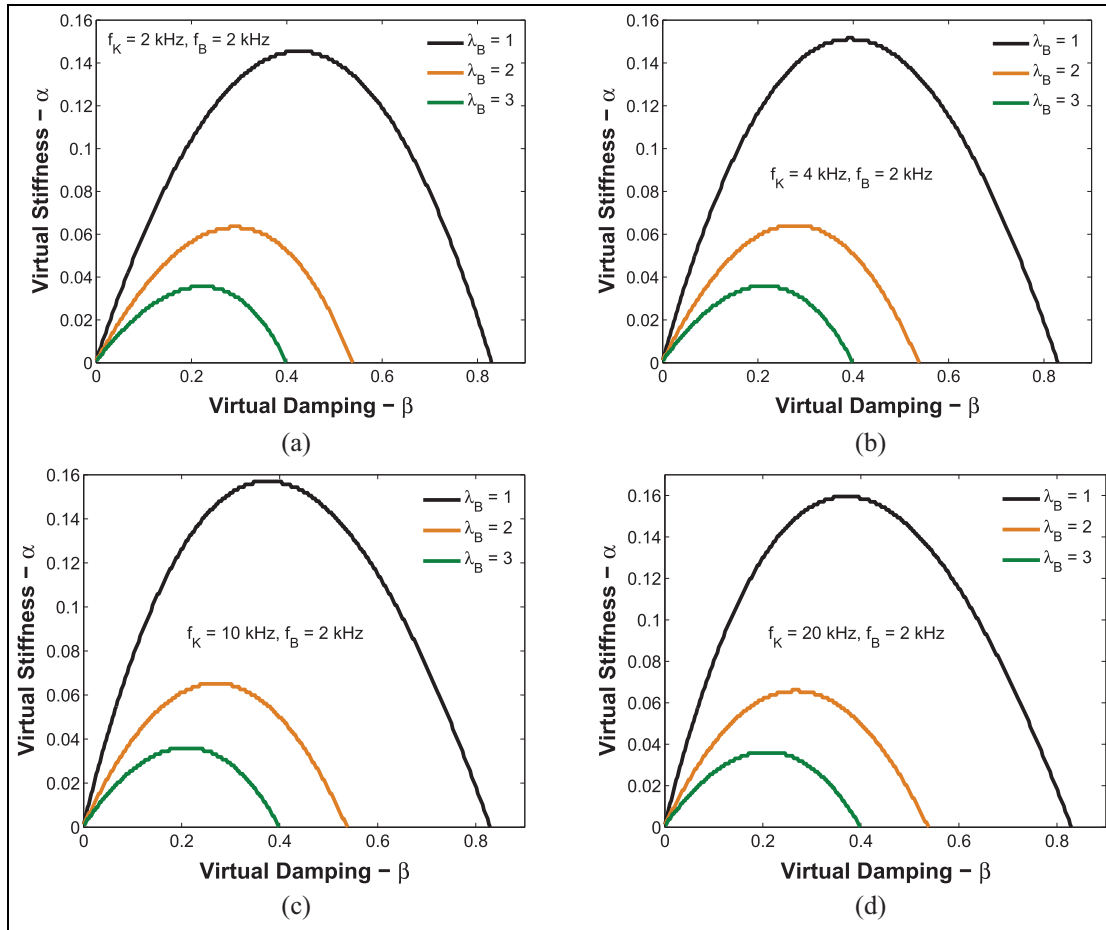


Figure 3. Stability boundaries (Non-dimensional Z-width plots) in the $\alpha - \beta$ plane for the 2-DOF haptic device under the dual-rate sampling scheme using the exact discrete-time stability formulation. The stability boundaries have been obtained for $T_B = 0.5$ ms ($f_B = 2$ kHz) and different values of f_K corresponding to $N = \{1, 2, 5, 10\}$. The values of λ_B correspond to time delays of $\lambda = \{0.5, 1, 1.5\}$ ms: (a) non-dimensional Z-width plots for $f_K = 2$ kHz, $f_B = 2$ kHz ($N = 1$), and $\lambda_B = \{1, 2, 3\}$, (b) non-dimensional Z-width plots for $f_K = 4$ kHz, $f_B = 2$ kHz ($N = 2$), and $\lambda_B = \{1, 2, 3\}$, (c) non-dimensional Z-width plots for $f_K = 10$ kHz, $f_B = 2$ kHz ($N = 5$), and $\lambda_B = \{1, 2, 3\}$, and (d) non-dimensional Z-width plots for $f_K = 20$ kHz, $f_B = 2$ kHz ($N = 10$), and $\lambda_B = \{1, 2, 3\}$.

outcomes arises from several factors. One such important factor is the assumption of rigid links in the device mechanism, whereas in practical scenarios, the actual device usually exhibits some structural flexibility, impacting the stability limits. This is particularly the case for haptic devices that employ capstan drives for transmission, as is the case in Koul.²⁷ As the emphasis of this study was to analyze the stability aspects of multi-DOF haptic devices subjected to multi-rate control techniques for virtual environment implementation, thus the assumption of link rigidity was made for simplifying the analysis. However, it is important to note that some prior works have utilized a two-mass model to capture the structural flexibility of the device.^{10,69} Moreover, the conservatism can be attributed to various phenomena, such as quantization, dry friction, motor saturation, unilateral constraints, etc., which are inherent in actual haptic interactions. These aspects were not modeled in this analysis due to their nonlinear nature and the difficulty associated with their modeling.^{17,19}

The stability limits, dynamics, force transmissibility, etc., of a multi-DOF haptic device are known to be influenced by its mechanical design. Previous research^{10,52,55} has explored the impact of the mechanical design on dynamic behavior, stability, and various performance metrics such as dexterity, manipulability, and force displayability under the uniform-rate control scheme. The theoretical analysis presented in previous sections considered a congruent pantograph mechanism with equal link lengths ($a = 10$ cm) and zero base length. However, for the purpose of investigating the combined effects of the dual-rate sampling scheme and mechanical design on the stability limits, particularly the maximum renderable stiffness, of a multi-DOF haptic device, two potential design variations have been considered. In the first case, the link length of the device is treated as a variable, with the base length assumed to be zero. In the second case, the base length is allowed to vary while keeping the link length constant. Following,⁵⁹ three configurations of the device were considered, as

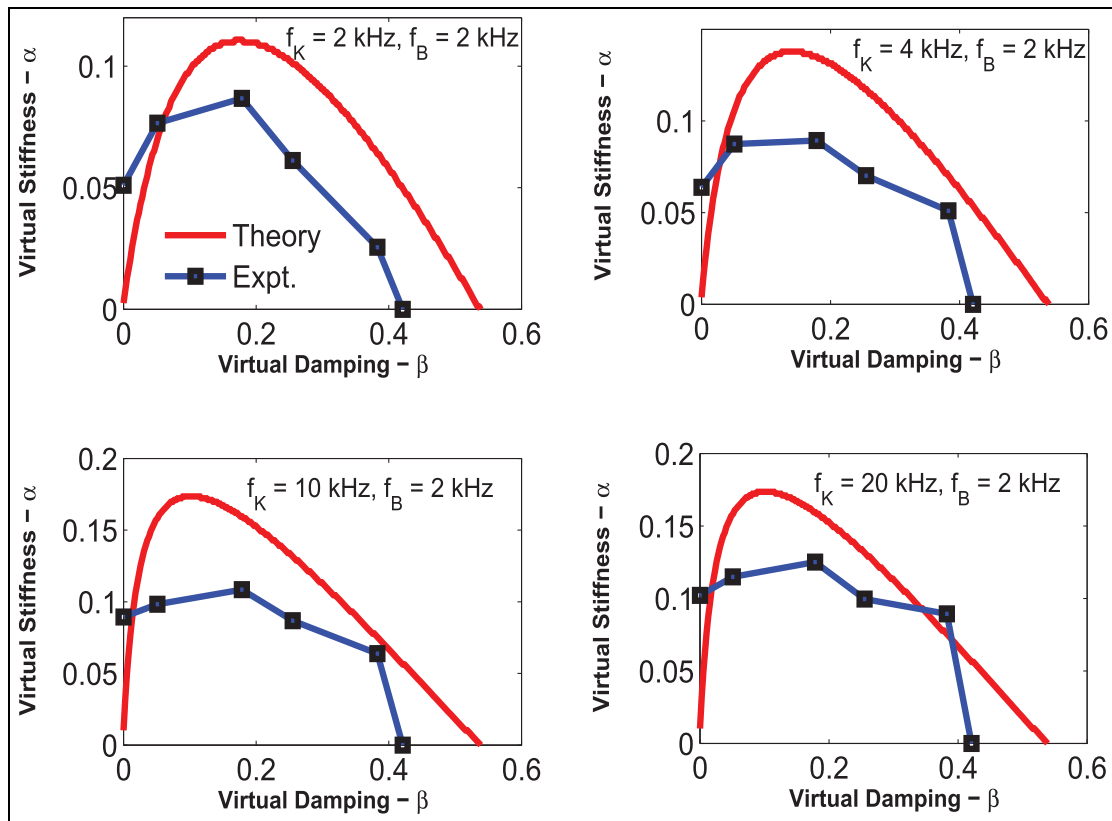


Figure 4. A comparison of the theoretical results with the experimental results of Koul.²⁷

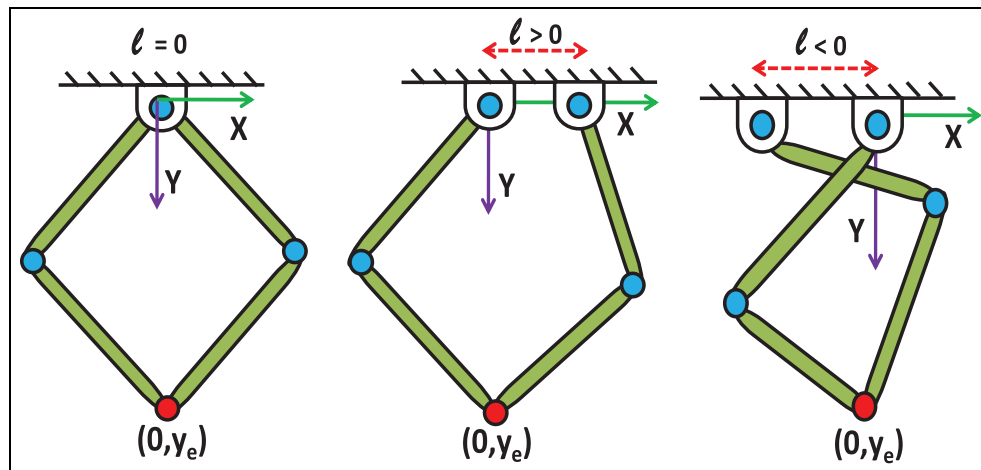


Figure 5. Schematic of different configurations of the 2-DOF device for determining the effect of ground link length ℓ on the stability limits of the dual-rate controller.

illustrated in Figure 5. While the first configuration with zero base length has already been examined in earlier section, the other two configurations can also be devised. These two configurations differ in the relative positions of the two active links. Depending on the relative positions of the active links, the base length value can be either positive or negative, as depicted in Figure 5. These two configurations are referred to as normal and crossed configuration,

respectively. The results of this investigation for a specific end-effector position $\mathbf{X} = [0.145]^T$ and $\lambda_B = 2$, and for two distinct dual-rate sampling combinations are depicted in Figure 6. An increase in the link length can be seen to be associated with an increase in the maximum virtual stiffness that the 2-DOF device can render at a particular operational-space position. Although this comparison is illustrated for only two sampling rate combinations,

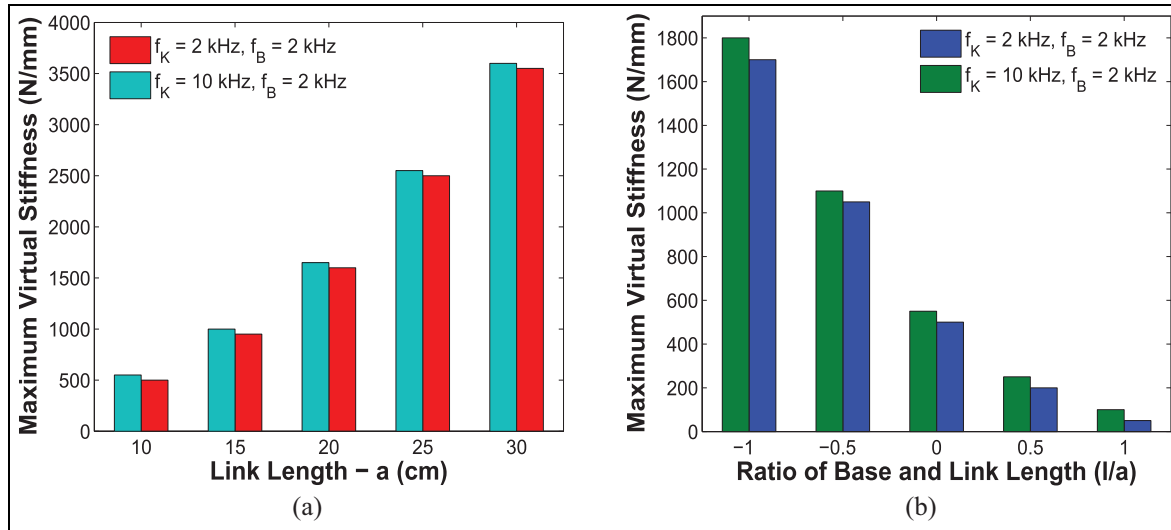


Figure 6. Effect of the device link length and base length on the maximum virtual stiffness for the 2-DOF haptic device subjected to dual-rate sampling scheme: (a) maximum virtual stiffnesses corresponding to different link lengths and zero base length, and (b) maximum virtual stiffnesses corresponding to different base lengths and link length of 10 cm.

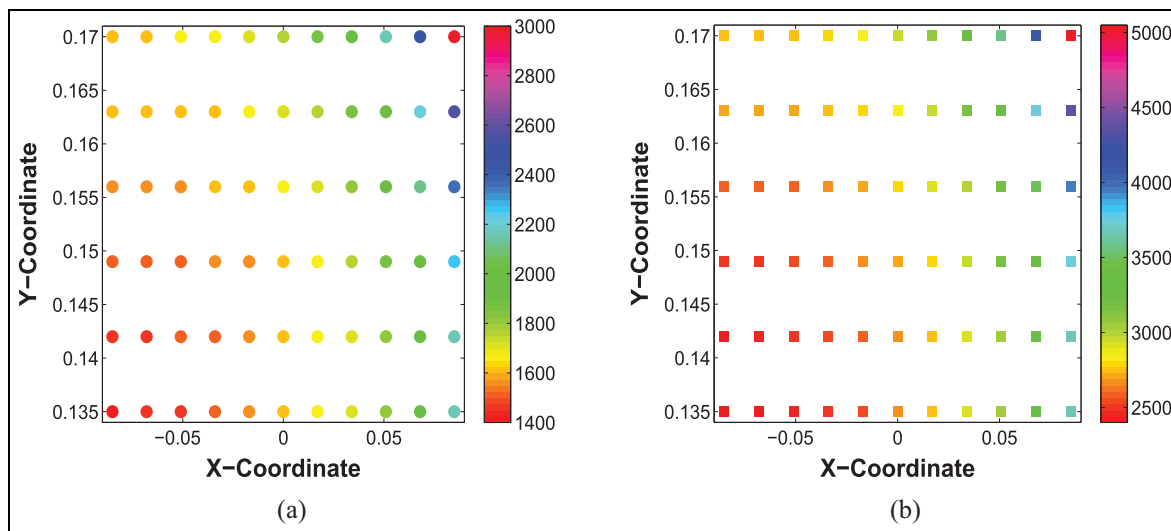


Figure 7. Variation of maximum virtual stiffness within the usable workspace of the 2-DOF haptic device subjected to dual-rate sampling scheme: (a) maximum virtual stiffness at different workspace locations for $f_K = 2 \text{ kHz}$, $f_B = 2 \text{ kHz}$, and $\lambda_B = 0$, and (b) maximum virtual stiffness at different workspace locations for $f_K = 10 \text{ kHz}$, $f_B = 2 \text{ kHz}$, and $\lambda_B = 0$.

similar results were obtained for other combinations as well. This increase in the stability limits is attributed to the increase in the effective mass Λ_{yy} and decrease of the viscous damping b_{yy} , experienced at the devices end-effector accompanied with an increase in the link length. Furthermore, as the base length increases for configuration with $l > 0$, the maximum virtual stiffness decreases while for the configuration with $l < 0$ an increase in base length is accompanied by an increase of the maximum virtual stiffness.

Due to the configuration dependence of the mass Λ_{yy} and damping b_{yy} felt at the end effector, it is essential to note that the stability limits of a multi-DOF device not only depend on the combinations of

sample rates or time delays but also vary location-wise within the device's usable workspace. For instance, Figure 7 illustrates the variation of the maximum virtual stiffness $K_{y,max}$ at different locations within the usable workspace of the 2-DOF device under consideration. These values were obtained by creating an 11×6 grid within the usable workspace, with the X-coordinate varying from -0.085 to 0.085 m and Y-coordinate ranging from 0.135 to 0.17 m . For $f_B = f_K = 2 \text{ kHz}$, and $\lambda_B = 0$, the maximum virtual stiffness ($K_{y,max}$) increases from 1400 to 3000 units corresponding to workspace locations of $(-0.085 \text{ m}, 0.135 \text{ m})$ and $(0.085 \text{ m}, 0.17 \text{ m})$, respectively. Similarly for $f_B = 2 \text{ kHz}$, $f_K = 10 \text{ kHz}$, and $\lambda_B = 0$, $K_{y,max}$

increases from 2400 to 5050 units corresponding to workspace locations of $(-0.085\text{ m}, 0.135\text{ m})$ and $(0.085\text{ m}, 0.17\text{ m})$, respectively. Furthermore, it can be observed that for all considered workspace locations, an increase in f_k from 2 to 10 kHz for a constant $f_B = 2\text{ kHz}$ and time delay, results in an increase in $K_{y,\max}$ thereby establishing the advantage of the dual-rate sampling approach for high stiffness rendering. Similar results were obtained for other higher sample combinations as well.

Conclusions and future work

This paper presents an analysis of the stability of a multi-DOF haptic device under a dual-rate control scheme, utilizing decoupled dynamics formulation and exact discrete-time stability analysis. The investigation includes an examination of the impact of time delay on theoretical stability boundaries, showing good agreement with experimental results. Additionally, the combined influence of dual-rate sampling and mechanical device design, specifically the effects of link lengths and base length, is explored. The analysis reveals that doubling and tripling the link length can increase the maximum renderable virtual stiffness by nearly factors of 3 and 6, respectively, for combinations of 10 and 2 kHz sampling rates with a two-sample time delay. Furthermore, providing a base length equal to link lengths increases the maximum virtual stiffness by approximately a factor of 3.5 for the crossed configuration but reduces it by nearly a factor of 5 for the normal configuration. Moreover, increasing the sampling rate of position loop from 2 to 10 kHz results in an increase of maximum virtual stiffness of about 70% within the devices different workspace locations. Stability analysis of multi-DOF dual-rate haptic rendering while considering factors such as device compliance, sensor quantization, coulomb damping, non-integral time delay, etc., will be a subject of our future work. Additionally, our future research will focus on using a hybrid control theory-machine learning-based approach for stability analysis of dual-rate haptic controllers.

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